LAMMPS Benchmark*

Student: Rufino Haroldo Locon

International Laboratory for Supercomputer Atomistic Modelling and Multi-scale Analysis

Introduction

The main usage and the major reason of the existence of High Performnace Computing Systems (HPC), is the possibility of making complicated calculus and simulations in a vast field of scientific applications. From Molecular Dinamics to Machine Learning Models, the HPC Systems represents and indispensable tools for science research and development.

In this work we are trying to determin the performace of the LAMMPS package, wich name comes from the acronym **Large-scale Atomic/Molecular Massively Parallel Simulator** This package is used by scientists, who want to investigate the behavior of molecules in different states and conditions.

We start talking about the core infrastructure of the **"cHRARISMa"** HPC System, implemented at the HSE University.

This HPC System consist of 6 blocks formed by a Proccesor Type and a GPU, as we can see at official site. `https://hpc.hse.ru/en/hardware/hpc-cluster`

For our benchmark test, we defined the type-a nodes with Intel Xeon Gold 6152 / NVIDIA Tesla V100 Processor

*This text was written with LaTex

and GPU, type-c with Intel Xeon Gold 6240R / NVIDIA Tesla V100 Processor and GPU, type-d with Intel Xeon Gold 6248R Processor only.

Procedure

For simulations on water model TIP3P, the work is based on the guide written by Ph.D. Christopher O'Brien, that can be find in the next link: `https://sites.google.com/a/ncsu.edu/cjobrien/tutorials-and-guides/working-with-water-in-lar`

It implements a predefined cubic structure for water simulation and TIP3 Potential. All files and results can be find at the following repository: `https://github.com/hlocon1309/lammps_bench`

The objective is to obtain the performace of the simulation with different parameters like number of cores, type of processor and type of gpu. The following task demostrates those results.

Using the tip3p.lmp input file, we make a run with different number of MPI processes, and got the performance in ns/day,

Task 1

Runnning the benchmark with LAMMPS package on type-a block.

Benchmark on LAMMPS TIP3P Water Model			
MPI (Processor type-a)	Loop Time	Performance (ns/day)	Wall Time
1	360.494	0.006	0:06:00
2	194.553	0.011	0:03:14
4	107.61	0.020	0:01:47
8	53.7858	0.040	0:00:53
16	29.5773	0.073	0:00:29

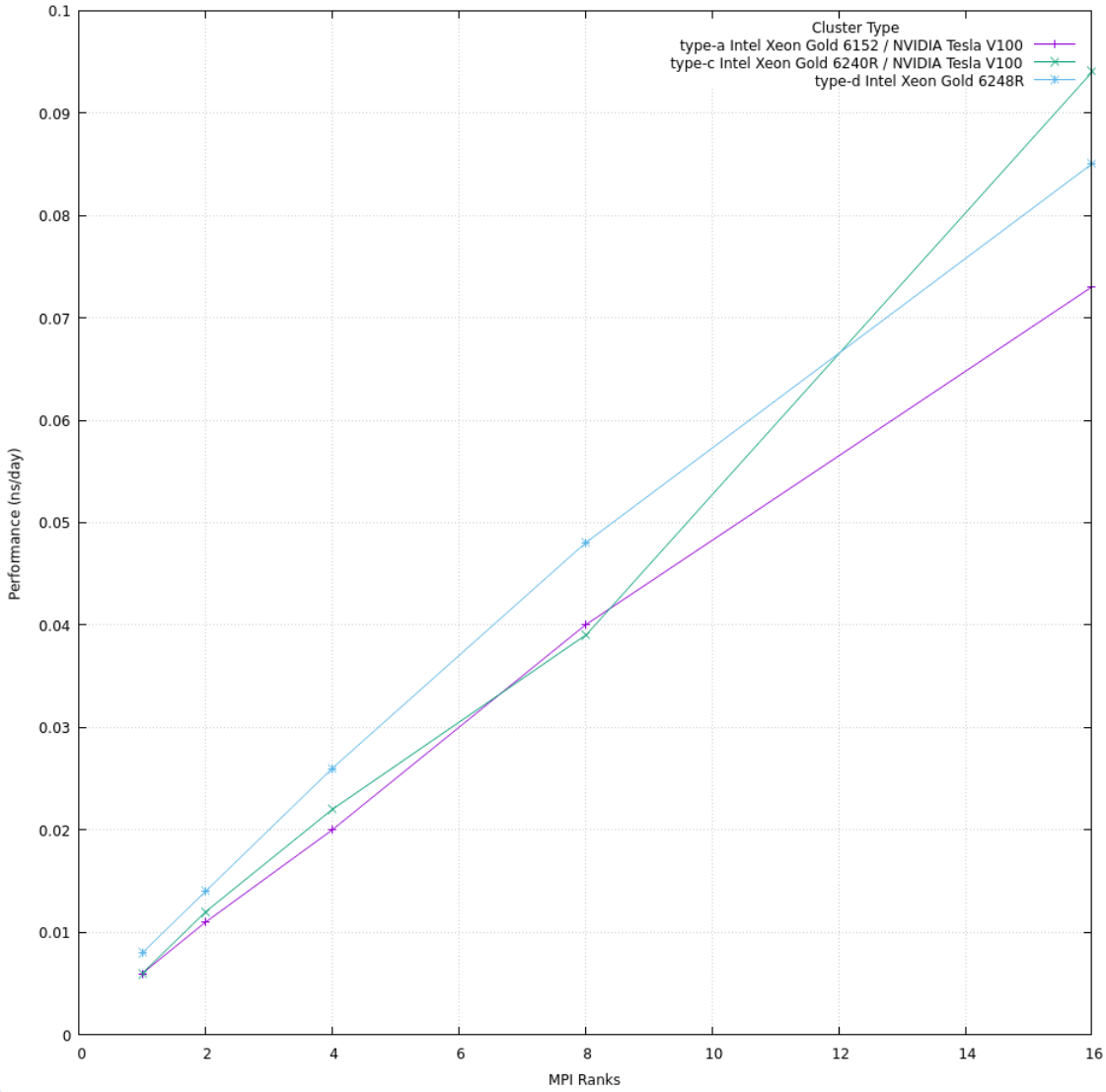
Task 2

Runnning the benchmark with LAMMPS package on type-c block.

Benchmark on LAMMPS TIP3P Water Model			
MPI (Processor type-c)	Loop Time	Performance (ns/day)	Wall Time
1	337.288	0.006	0:05:37
2	176.599	0.012	0:02:56
4	97.7465	0.022	0:01:37
8	54.752	0.039	0:00:54
16	22.8791	0.094	0:00:22

Task 3

Runnning the benchmark with LAMMPS package on type-d block.

Benchmark on LAMMPS TIP3P Water Model			
MPI (Processor type-d)	Loop Time	Performance (ns/day)	Wall Time
1	275.501	0.008	0:04:35
2	150.984	0.014	0:02:31
4	82.0875	0.026	0:01:22
8	45.0075	0.048	0:00:45
16	25.2986	0.085	0:00:25

LAMMPS Benchmark Graphic Result